

$$T(n) - 2T(n-1) = n2^n$$

Let

$$x(n) = T(n)u(n)$$

$$\Rightarrow x(0) = 1, \quad x(1) = T(1) \dots \text{and so on}$$

$$x(n) = 0 \quad \text{for } n < 0$$

$$\Rightarrow x(n) - 2x(n-1) = n2^n u(n) + \delta(n)$$

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Taking z-transform

$$X(z) - 2z^{-1}X(z) = Z\{n2^n u(n)\} + 1 \quad \text{--- (1)}$$

$$\therefore Z\{F(n)\} = \sum_{n=0}^{\infty} F(n)e^{-n}$$

$$\Rightarrow Z\{na^n u(n)\} = Z\{na^n\} = \frac{az^{-1}}{(1-az^{-1})^2}$$

Put this in (1)

$$\Rightarrow X(z) = \frac{1 + \frac{2z^{-1}}{(1-2z^{-1})^2}}{1-2z^{-1}}$$

After simplify

$$X(z) = \frac{1 - 2z^{-1} + 4z^{-2}}{(1-2z^{-1})^3}$$

By partial fractⁿs,

$$\Rightarrow X(z) = \frac{1}{1 - 2z^{-1}} - \frac{1}{(1 - 2z^{-1})^2} + \frac{1}{(1 - 2z^{-1})^3}$$

Take inverse

$$\Rightarrow x(n) \Rightarrow \text{~~the result~~} = 2^n u(n) - (n+1)2^n u(n) + \frac{(n+1)(n+2)}{2} 2^n u(n)$$